

Week 1

Team project update/progress

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Weekly Progress Summary

GitHub	Team familiarized with basic concepts
Documentation	Reviewed and analyzed Code/PDF documentation
Architecture	Studied agent roles and system architecture
Onboarding	Team discussed about the UI design scope and objectives
UI Design	Explored design options and created prototypes
Selection	Compared prototypes and selected final design

GitHub Familiarization

Repository

Centralized file and code management

Version Control

Change tracking and version history

Task Division

Issue-based work assignment and tracking

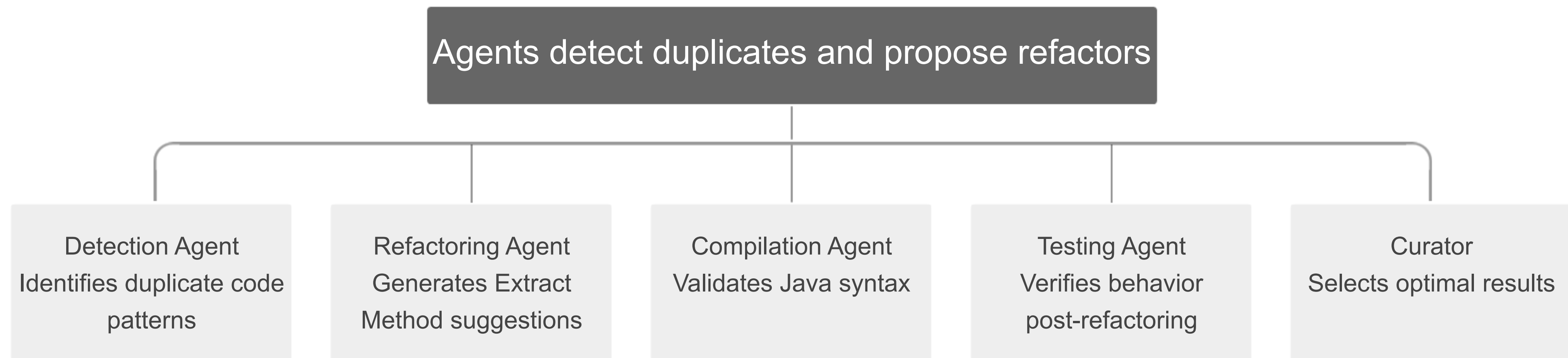
Peer Review

Pull request workflow for code review

Git Commands

clone, pull, add, commit, push, branch, PR

Multi-Agent System Architecture



Detection Agent Workflow

Refactoring Agent Workflow

Compilation and Testing Agents

Target Audience & Requirements

Target Users

Java developers and students seeking code quality improvement

Tool Characteristics

Clear, fast, and trustworthy with minimal workflow disruption

Detection Display

Show detected code with file location and line numbers

Change Visualization

Display proposed refactoring changes with compilation and testing validation status

User Actions

Apply refactoring, Edit suggestions, View Diff comparison, or Reject changes



The tool empowers developers to improve code quality efficiently while maintaining full control

Prototype Evaluation and Selection

Prototype Creation

Team created multiple design prototypes for side-by-side comparison

Clarity Assessment

Evaluated usability and visual clarity across all design options

Detail Communication

Assessed how well each design communicates and displays information

Final Selection

Chose design that best displays clone type, location, diff preview, and validation

Roadmap and Next Steps

Design Refinement

Polish and finalize the selected prototype design

GitHub Organization

Start utilizing it and working on branches

UI Implementation

Advance front-end development of the tool interface

Progress Tracking

Establish a website to monitor and communicate project progress